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Synopsis on

AI Based Automated Inspection and Defect Detection for Railway Safety

Submitted in partial fulfilment of the requirements for the Award of degree of

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Introduction:

Railway transportation is one of the most widely used and essential modes of transport, playing a crucial role in the movement of passengers and goods. Ensuring railway safety is therefore of utmost importance, as even minor defects in critical components can lead to serious accidents. Among these, defects in railway wheels and undercarriage components such as cracks, shelling, and surface wear pose significant risks, as they develop gradually due to continuous mechanical stress and harsh operating conditions. If not detected at an early stage, these defects can result in derailments, equipment failure, and costly disruptions to railway operations. Traditionally, inspection of such components has been carried out manually, which is time-consuming, labour intensive, and often prone to human error. The increasing demand for safer and more reliable railway systems highlights the need for continuous monitoring and early detection of faults to prevent accidents and ensure smooth operation.

Literature Survey:

Recent advancements in railway inspection systems have increasingly leveraged artificial intelligence and machine learning techniques for automated defect detection. Thakkar et al. [1] explored anomaly detection methods for vehicle health monitoring using techniques such as Mahalanobis distance and autoencoders, highlighting the importance of threshold selection. Nikitha et al. [2] proposed an IoT-based railway track monitoring system using sensors and cameras for real-time detection; however, the approach depends heavily on hardware deployment, limiting scalability.

Renatus et al. [3] reviewed perception systems in automated railway transport using sensors such as LiDAR and cameras, emphasizing robust fault detection mechanisms. Kirubagaran and Nagarajan [4] utilized acoustic signal analysis combined with machine learning models for crack detection, though reliance on synthetic data affects reliability. Deng et al. [5] introduced an enhanced YOLOv8 model for railway hazard detection, achieving improved accuracy but facing challenges in real-world generalization.

Deep learning approaches have shown promising results, as demonstrated by Amritsai et al. [6], who evaluated CNN models such as ResNet and DenseNet for defect detection. Tao et al. [7] used PS-InSAR techniques for anomaly detection, while Zhang et al. [8] proposed a multimodal model combining visual and textual data. Other studies include LiDAR-based detection [9], systematic reviews of machine vision methods [10], and AI-based monitoring techniques [11–15].

Overall, existing research highlights the effectiveness of AI-based railway inspection systems but also reveals limitations such as dependency on specialized hardware, lack of standardized datasets, and limited real-time implementation.

Research Gap:

Railway wheels may develop defects such as cracks and surface damage during operation, and detecting these defects in moving trains is difficult using traditional manual inspection methods. Manual inspection is often time-consuming, inconsistent, and may fail to identify early-stage defects. Therefore, there is a need to develop a reliable automated inspection system

using AI along with proper camera calibration techniques to accurately detect defects in railway wheels from image and video data.

Problem Statement:

Railway safety requires an AI-based automated inspection system for detecting defects in critical components such as wheels. Existing manual inspection methods are time-consuming, error-prone, and inefficient for real-time monitoring. Therefore, there is a need for an automated system that can analyze image and video data to accurately detect railway wheel defects and enhance safety.

Objectives:

- To design an AI-based automated inspection system for railway wheel defect detection.
- To collect and preprocess image and video data of railway wheels.
- To implement camera calibration techniques for accurate image analysis and synchronization.
- To train and evaluate a deep learning model for detecting cracks and defects in railway wheels.
- To develop a system capable of analyzing frames extracted from video and identifying defective components.
- To improve railway safety by enabling early detection of structural faults.

Hardware Requirements:

- High-resolution cameras or video dataset of railway wheels
- Computer system with sufficient processing capability

Software Requirements:

- Python programming language
- OpenCV (for image processing and video frame extraction)
- Roboflow (for dataset annotation and management)
- YOLO-based deep learning framework (Ultralytics YOLO)
- PyTorch / TensorFlow for model training

Methodolgy:

- The overall workflow of the proposed system is illustrated in Fig 1. Initially, camera calibration is performed to synchronize cameras and correct image distortion.
- Collect images and videos of railway wheels.
- Extract frames and preprocess the data.
- Detect and isolate the wheel region from frames.
- Train a deep learning model to identify cracks and defects.
- Analyze video frames and detect defective wheels automatically.
- Evaluate results.

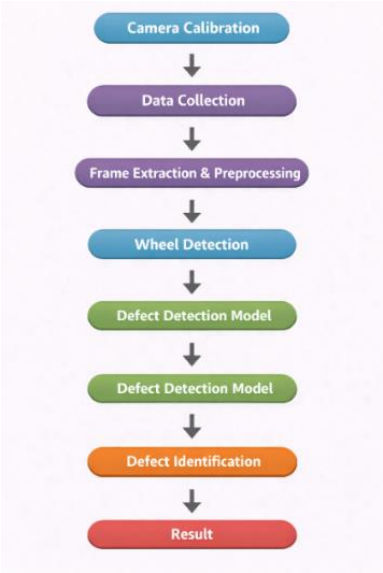

Fig 1: Proposed System Architecture for AI-Based Railway Wheel Defect Detection

Expected Outcomes:

- Development of an AI-based system capable of detecting defects such as cracks in railway wheels from images and video data.
- Accurate identification of defects with good performance metrics (precision, recall, and mAP).
- Implementation of camera calibration to improve image quality and detection accuracy.
- Creation of a processed and annotated dataset of railway wheel images.
- Ability to analyze video frames and automatically identify defective components.
- Improved reliability and efficiency compared to manual inspection methods.

Limitations:

- Limited dataset availability for real-world railway defect scenarios
- Difficulty in capturing complete wheel surface due to motion and camera angle constraints
- Sensitivity to lighting conditions, motion blur, and environmental noise
- Current system focuses only on visible defects and may not detect internal faults
- Computational requirements for training deep learning models

Enhancement Opportunities:

- Integration of real-time detection systems using live camera feeds
- Expansion to detect defects in other components such as axles and suspension systems
- Use of multi-camera setups for better coverage of wheel surfaces
- Implementation of edge computing devices for on-site processing
- Incorporation of predictive maintenance systems using historical data
- Use of advanced models for improved accuracy and segmentation-based detection

Conclusion:

This project presents an AI-based automated inspection system for detecting defects in railway wheels using image and video analysis. By leveraging deep learning techniques and computer vision, the system aims to improve the accuracy and efficiency of defect detection compared to traditional manual methods. The integration of camera calibration further enhances the reliability of the system. The proposed approach has the potential to contribute to safer railway operations by enabling early detection of defects and reducing the risk of accidents.

Bibliography:

[1]K. Thakkar, N. Hemachandra, and T. R. Singh, *"Anomaly Detection Methods: Application to Automated Vehicle Health Monitoring,"* in *Data Science for Modeling Managerial and Socioeconomic Problems*, Springer, 2026, pp. 391–425.

[2] R. Nikitha, M. Nishchitha, and M. Vismaya, *"Railway Track Monitoring for Obstacles and Damage Detection using AI and ML," Int. J. Adv. Res. Sci. Commun. Technol.*, vol. 108, 2025.

[3] K. Renatus et al., *"Fault detection, isolation, identification and recovery methods for perception systems in automated railway transport: A review," Transportation Engineering*, vol. 23, 2026.

[4] K. Kirubagaran and D. R. Nagarajan, *"Railway Track Crack Detection Using Sound Signal Analysis and Gen AI Models," IJSREM*, 2026.

[5] R. Deng et al., *"Enhanced YOLOv8 for multi-hazard detection on railways using hybrid datasets," Measurement Science and Technology*, 2026.

[6] S. Amritsai et al., *"Performance analysis of deep convolutional neural network models for railway track damage detection," Engineering Research Express*, vol. 8, 2026.

[7] R. Tao et al., *"Monitoring of ground displacement-induced railway anomalies using PS-InSAR techniques," Measurement*, vol. 248, 2025.

[8] J. Zhang et al., *"Railway-CLIP: A multimodal model for abnormal object detection," High-speed Railway*, vol. 3, no. 3, 2025.

[9] Y. Yan et al., *"Advancing Railway Infrastructure Monitoring: Railway Pole Detection," Computers, Materials and Continua*, vol. 83, no. 2, 2025.

[10] A. Kumar and S. P. Harsha, *"A systematic literature review of defect detection in railways using machine vision-based methods," Int. J. Transportation Science and Technology*, vol. 18, 2025.

[11] A. Khajehdezfuly et al., *"Applications of AI in railway monitoring and maintenance," Journal of Industrial Information Integration*, vol. 48, 2025.

[12] S. Fathi et al., *"Hybrid ANN–GA technique for anomaly detection in railway substructure," ICE Transport*, vol. 178, no. 6, 2024.

[13] R. Ghiasi et al., *"Unsupervised anomaly detection using one-class SVM for railway defects," Engineering Applications of AI*, vol. 133, 2024.

[14] M. Heinrich et al., *"Rule-based anomaly detection for railway signalling networks," Int. J. Critical Infrastructure Protection*, vol. 42, 2023.

[15] F. Li et al., *"Rail-PillarNet: A 3D detection network for railway objects using LiDAR," Computers, Materials and Continua*, vol. 80, no. 3, 2024.